The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5

Max Marks: 100

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Course: Open Source Software

Your choice	Write your chosen software here before submission: _____

My choice :

Git	Version Control	GPL v2	*The tool Linus built when proprietary failed him*

INTRODUCTION

In the world we live in today software is used in every area of life like education, business, healthcare and communication. Most of these systems need software that works well and does not fail. One of the ideas in making software is Open Source Software.

Open Source Software lets people look at the code study it make changes and share it with others without any problems. This is different from software that companies own where the code's secret and only the company can control it.

For this project I wanted to learn about open source ideas both by reading about them and, by trying them out. I chose Apache HTTP Server as the software to look at. Apache is one of the used web servers and it helps host websites on the internet.

The main things I wanted to do in this project are:

- Learn where open source came from and what it is about
- Look at the rules and ethics of open source
- See how the software works on Linux
- Explore the community around it
- Compare it to software that is owned by companies

I also made shell scripts to learn about Linux commands and how to automate tasks. These scripts helped me understand how to use commands to automate work.

This project helped me understand how Open Source Software works in the world. I now know how Open Source Software is used in systems. Open Source Software is very important. It is used in many areas of life.

The Open Source Audit

Part A — Origin and Philosophy

A1 — Origin Story (Apache HTTP Server)

The Apache HTTP Server was created in the 1990s. Then the internet was growing really fast. People needed a web server that was stable and could handle lots of traffic.

The Apache HTTP Server was what people needed. Before it developers used a server called NCSA HTTPd.. The NCSA HTTPd server wasn't getting the support it needed. So a group of developers started working on it by adding patches. Over time all these patches became the Apache HTTP Server.

People often say the Apache HTTP Server got its name because it was like a patchwork server. This is true. The Apache HTTP Server was made from all the patches that the developers added.

The goal of the Apache HTTP Server was simple. It was supposed to do a things.

- Create a web server that people could rely on
- Make the Apache HTTP Server open-source so anyone could use it for free
- Let developers change the Apache HTTP Server and make it better

The Apache HTTP Server solved some problems.

- It let many users access the web without slowing down
- It handled web traffic well so it didn't get overwhelmed
- It gave developers the freedom to make changes to the Apache HTTP Server

Today the Apache HTTP Server is one of the popular web servers in the world. Many websites use the Apache HTTP Server. This shows how working together on open-source

projects, like the Apache HTTP Server can create software. The Apache HTTP Server is an example of this.

A2 — License Analysis (Apache License 2.0)

Apache HTTP Server is released under the Apache License 2.0. This is a license that lets people use, change and share the software easily.

Here are some key things users can do:

- Use the software for anything they want
- Change the code to make it work
- Share the original or changed code with others

The Apache License is different from GPL. With GPL if you change the code you have to share those changes with everyone.

With Apache License you can make changes and keep them private.

Another good thing about the Apache License is that it protects users from patent problems.

Here's how Apache License and GPL compare:

- GPL says you must share any changes you make
- Apache License lets you make changes and keep them to yourself

This makes Apache License good, for both people and businesses.

In short the Apache License gives users freedom and flexibility to use and change the Apache HTTP Server..

A3 — Ethics of Open Source

Open source is not about technology it is also about doing what is right.

It helps with:

- Sharing what we know

- Working
- Being honest and open

Anyone can look at open-source code and make it better. This helps create a group of people from all around the world who work together.

There are some things that people do not agree on:

- Should all software be open source?
- Is it okay for companies to make money from open-source tools?

A lot of companies use open-source tools. Build new things using them.. They also help make the software better.

The idea of using the work of people who came before us is really important here. Open source developers build on what's already there instead of starting from nothing.

In my opinion open source is a way to learn new things and come up with new ideas at the same time. Open source creates a balance, between learning from open source and creating new things with open source.

Part B — Linux Footprint of Apache

Open source matters for a lot of reasons. It is not about the technology it is also about doing what is right.

For example Apache HTTP Server is something that people can use for free. This is really good because people do not have to pay any money to use Apache HTTP Server.

If you have a computer that runs Linux you can install Apache HTTP Server on it. This is easy to do.

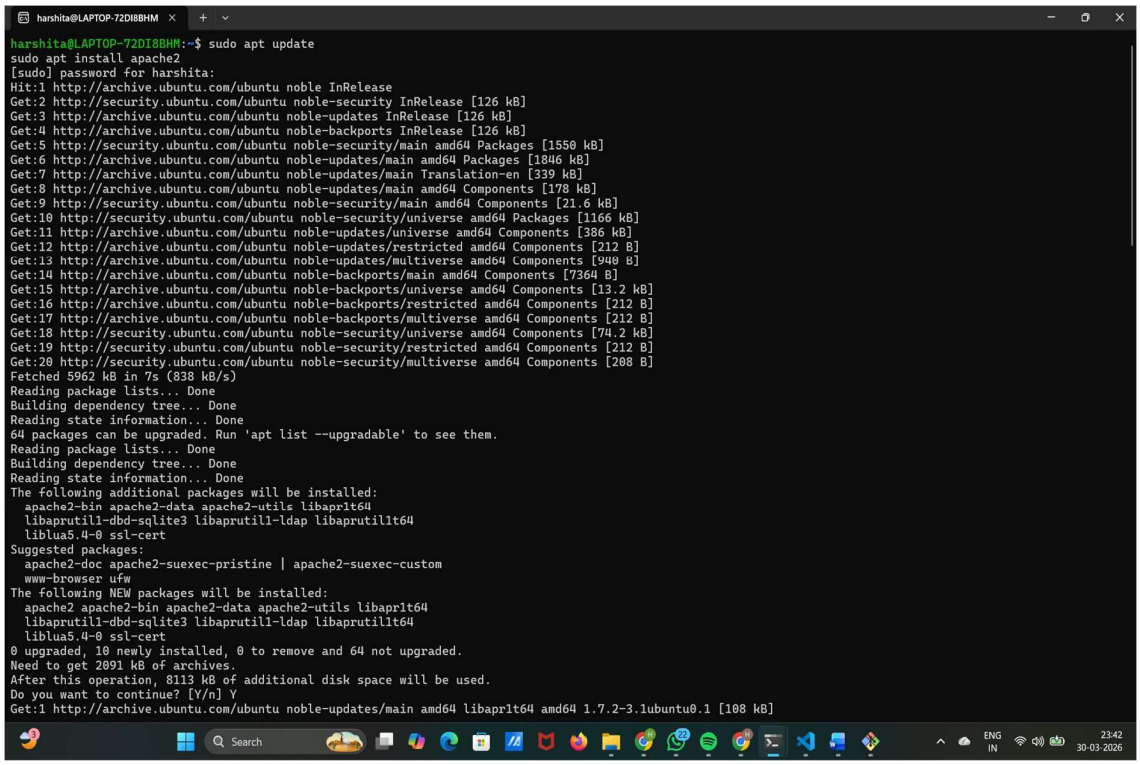
You can do this using a tool called the package manager. The package manager is really easy to use.

All you need to do is open the terminal and type one line of text. This line of text is not hard to remember.

To get Apache HTTP Server, you type: sudo apt install apache2. This line of text tells the computer to get the Apache package, from Linux websites and install it on your computer.

When you type this line of text you get Apache HTTP Server. You use Apache HTTP Server on your Linux computer.

Apache HTTP Server is what you get when you use the package manager to install it. The package manager helps you get Apache HTTP Server on your Linux computer.



```
harshita@LAPTOP-72DI8BHM:~$ sudo apt update
sudo apt install apache2
[sudo] password for harshita:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1550 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1846 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1166 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7364 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Fetched 5962 kB in 7s (838 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
64 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom
  www-browser ufw
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 64 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
```

Figure 1: Apache installation command showing required packages

```
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../4-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../5-apache2-bin_2.4.58-1ubuntu8.11_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu8.11) ...
Selecting previously unselected package apache2-data.
Preparing to unpack .../6-apache2-data_2.4.58-1ubuntu8.11_all.deb ...
Unpacking apache2-data (2.4.58-1ubuntu8.11) ...
Selecting previously unselected package apache2-utils.
Preparing to unpack .../7-apache2-utils_2.4.58-1ubuntu8.11_amd64.deb ...
Unpacking apache2-utils (2.4.58-1ubuntu8.11) ...
Selecting previously unselected package apache2.
Preparing to unpack .../8-apache2_2.4.58-1ubuntu8.11_amd64.deb ...
Unpacking apache2 (2.4.58-1ubuntu8.11) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../9-ssl-cert_1.1.2ubuntu1_all.deb ...
Unpacking ssl-cert (1.1.2ubuntu1) ...
Setting up ssl-cert (1.1.2ubuntu1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ssl-cert.service → /usr/lib/systemd/system/ssl-cert.service.
Setting up libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Setting up liblua5.4-0:amd64 (5.4.6-3build2) ...
Setting up apache2-data (2.4.58-1ubuntu8.11) ...
```

Figure 2: Installation process of Apache packages

```
Setting up apache2-data (2.4.58-1ubuntu8.11) ...
Setting up libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Setting up apache2-utils (2.4.58-1ubuntu8.11) ...
Setting up apache2-bin (2.4.58-1ubuntu8.11) ...
Setting up apache2 (2.4.58-1ubuntu8.11) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
harshita@LAPTOP-72DI8BHM:~$
```

Figure 3: Successful installation and configuration of Apache server

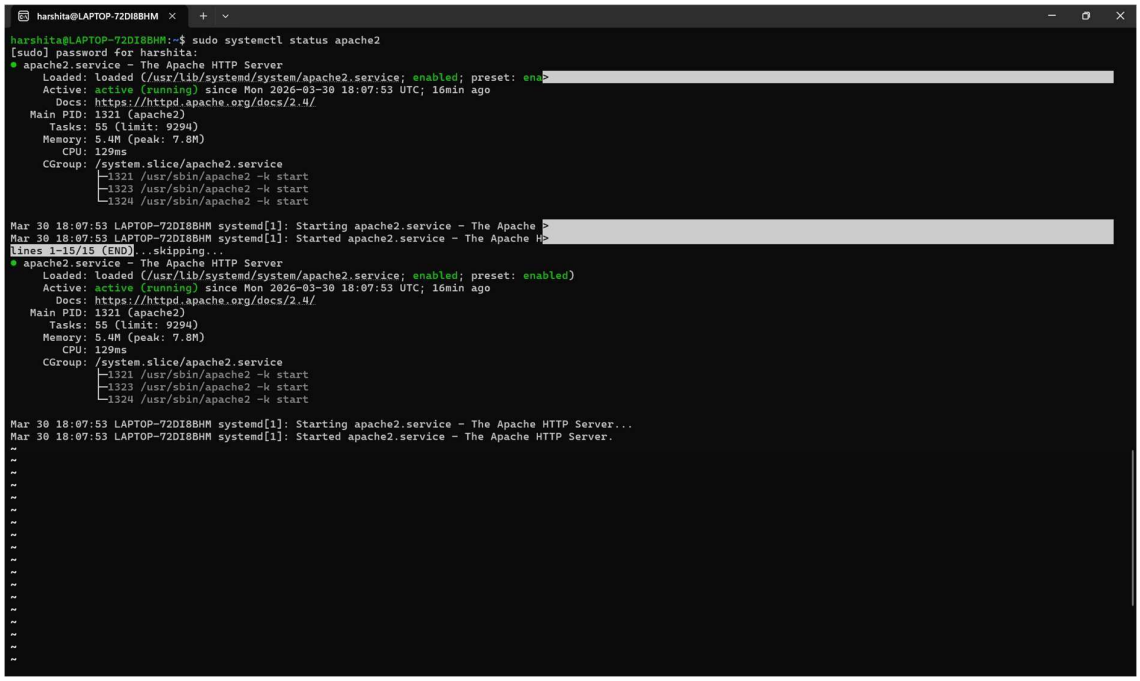


Figure 4: Status of Apache service showing that the server is active and running

The above output shows that the Apache web server is currently active and running. This confirms that the installation was successful and the service is functioning properly.

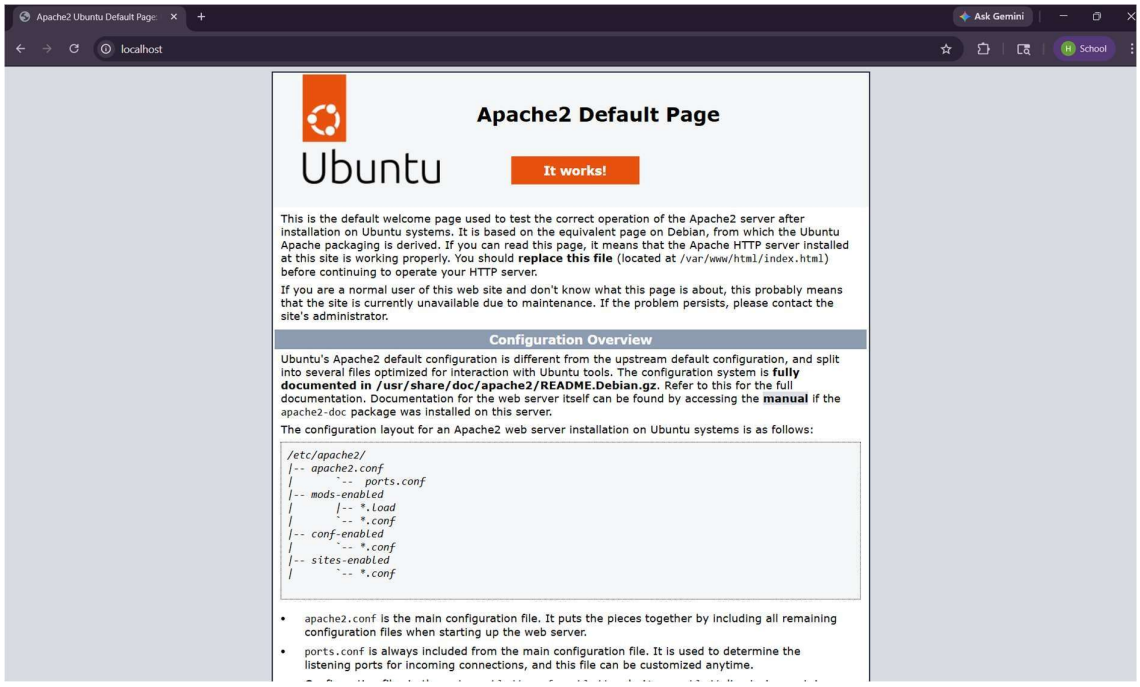


Figure 5: Apache default web page displayed on localhost after successful installation

The above screenshot shows the default Apache web page opened in the browser using the localhost address. This confirms that the Apache server is successfully installed and running, and it is able to handle web requests properly.

Part C — The FOSS Ecosystem Around Apache

Apache is not something you can use on its own. It is part of a group of open source tools. Apache works with other tools and technologies to build complete web applications.

Apache is mostly written in the C programming language. It needs the Linux operating system to work. It also uses things like APR, which's Apache Portable Runtime to make sure it runs well.

These things help Apache be fast not use up much space and be reliable.

Apache is an important part of the LAMP stack, which is made up of:

- Linux. This is the operating system
- Apache. This is the web server
- MySQL. This is the database
- PHP. This is used for server side scripting

All these things work together to make websites and web applications that can change.

Lots of platforms and technologies use Apache. It is very common in web hosting. Can work with many programming languages like PHP, Python and Perl.

Apache also works with things like Docker and cloud platforms.

The people who make Apache have a community. Developers from over the world help out by fixing problems adding new things and making it run better.

The Apache Software Foundation is in charge of making sure Apache is good and safe.

In conclusion Apache is a part of the open source ecosystem. It works with tools and technologies to provide complete web solutions. Apache has a strong community and lots of people use it which makes it one of the reliable web servers. Apache is very important, for making web applications. Apache helps make sure they run well.

Part D — Open Source vs Proprietary Software

Open-source software Open-source software and proprietary software are two ways of making software.

- One is open-source software.

It lets users see and change the code.

- The other is software.

It keeps the code secret. A company controls it.

Open-source software is different, from software. People can. Change open-source software. Proprietary software is only used by people who buy it. It is made by a company. The company owns it. Users do not own it. They only use it.

Feature	Apache (Open Source)	Proprietary Alternative (e.g., Perforce)
Cost	Free to use	Paid license required
Source code	Open and accessible	Closed and restricted
Flexibility	High	limited
Support	Community-based	Modification is restricted
Feature	Apache (Open Source)	Proprietary Alternative (e.g., Perforce)

Control	Shared with community	Controlled by company

Open source software like Apache is really good because it is flexible. You can see what is going on. You can change the software to do what you want. There are people who can help you.On the hand proprietary software is good too because it has good support and it works well.. You do not have control over it and you have to pay to use it.

Open source software like Apache is better, for people who are learning and trying to make new things because it is easy to get and you can do what you want with it.. Big companies might like proprietary software better because they need help from the people who made it and they are willing to pay for it.

SCRIPTS

Script 1 — System Identity Report

This script is used to display basic system information such as the current user, hostname, kernel version, uptime, and current date. It uses command substitution to store the values in variables and prints them using formatted output. This script helps in understanding how system information can be accessed using simple Linux commands and variables in shell scripting.

```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ cat script1_system_identity.sh
#!/bin/bash
# ============================================================
# Script 1: System Identity Report
# Author: [Your Name] | Registration: 24BAI10405
# Course: Open Source Software | VIT
# Description: Displays a welcome screen with Linux system info
#              relevant to the Apache HTTP Server OSS audit.
# ============================================================

# --- Student & Project Variables ---
STUDENT_NAME="Harshita"          # Replace with your full name
REG_NUMBER="24BAI10405"
SOFTWARE_CHOICE="Apache HTTP Server"
SOFTWARE_LICENSE="Apache License 2.0"

# --- Gather System Information using command substitution ---
KERNEL=$(uname -r)                          # Linux kernel version
DISTRO=$(cat /etc/*release 2>/dev/null | grep -m1 "PRETTY_NAME" | cut -d= -f2 |
tr -d '"')
# Fallback if PRETTY_NAME not found
[ -z "$DISTRO" ] && DISTRO=$(uname -o)

USER_NAME=$(whoami)                          # Current logged-in user
HOME_DIR=$HOME                               # Home directory of current user
UPTIME=$(uptime -p 2>/dev/null || uptime)  # Human-readable uptime
CURRENT_DATE=$(date '+%A, %d %B %Y')       # e.g. Monday, 01 January 2025
CURRENT_TIME=$(date '+%H:%M:%S')           # e.g. 14:30:00
HOSTNAME=$(hostname)                         # Machine hostname

# --- Check if Apache is installed (cross-distro) ---
# Ubuntu/Debian use 'apache2', CentOS/RHEL use 'httpd'
if command -v apache2 &>/dev/null; then
    APACHE_VERSION=$(apache2 -v 2>/dev/null | grep "Server version" | awk '{prin
t $3}')
elif command -v httpd &>/dev/null; then
    APACHE_VERSION=$(httpd -v 2>/dev/null | grep "Server version" | awk '{print
$3}')
else
   APACHE_VERSION="Not installed"
fi

# --- Display the Identity Report ---
echo "============================================================"
echo "        OPEN SOURCE AUDIT – SYSTEM IDENTITY REPORT"
echo "============================================================"
echo ""
echo "  Student   : $STUDENT_NAME"
echo "  Reg No    : $REG_NUMBER"
echo "  Software  : $SOFTWARE_CHOICE"
echo ""
echo "------------------------------------------------------------"
echo "  SYSTEM INFORMATION"
echo "------------------------------------------------------------"
echo "  Hostname        : $HOSTNAME"
echo "  Distribution    : $DISTRO"
echo "  Kernel Version  : $KERNEL"
echo "  Logged-in User  : $USER_NAME"
echo "  Home Directory  : $HOME_DIR"
echo "  System Uptime   : $UPTIME"
echo "  Date            : $CURRENT_DATE"
echo "  Time            : $CURRENT_TIME"
echo ""
```

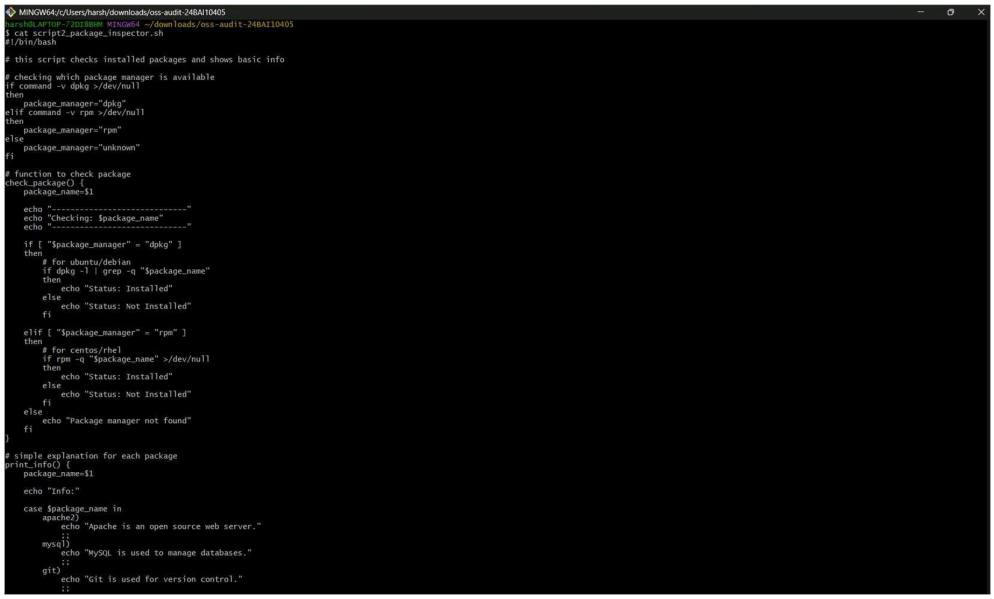
Bash script for displaying system identity

```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405
echo "  Logged-in User  : $USER_NAME"
echo "  Home Directory  : $HOME_DIR"
echo "  System Uptime   : $UPTIME"
echo "  Date            : $CURRENT_DATE"
echo "  Time            : $CURRENT_TIME"
echo ""
echo "------------------------------------------------------------"
echo "  OPEN SOURCE SOFTWARE DETAILS"
echo "------------------------------------------------------------"
echo "  Software        : $SOFTWARE_CHOICE"
echo "  License         : $SOFTWARE_LICENSE"
echo "  Apache Version  : $APACHE_VERSION"
echo ""
echo "  License Note:"
echo "  The Apache License 2.0 is a permissive free software license."
echo "  It allows anyone to use, modify, and distribute the software"
echo "  for any purpose, including commercial use, without requiring"
echo "  derivative works to be open-sourced (unlike the GPL)."
echo ""
echo "============================================================"
echo "  Learning Linux step by step is really helpful."
echo "============================================================"

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script1_system_identity.sh
script1_system_identity.sh: line 24: uptime: command not found
============================================================
        OPEN SOURCE AUDIT – SYSTEM IDENTITY REPORT
============================================================

  Student   : Harshita
  Reg No    : 24BAI10405
  Software  : Apache HTTP Server

------------------------------------------------------------
  SYSTEM INFORMATION
------------------------------------------------------------
  Hostname        : LAPTOP-72DI8BHM
  Distribution    : Msys
  Kernel Version  : 3.6.6-1cdd4371.x86_64
  Logged-in User  : harsh
  Home Directory  : /c/Users/harsh
  System Uptime   :
  Date            : Tuesday, 31 March 2026
  Time            : 01:16:45

------------------------------------------------------------
  OPEN SOURCE SOFTWARE DETAILS
------------------------------------------------------------
  Software        : Apache HTTP Server
  License         : Apache License 2.0
  Apache Version  : Not installed

  License Note:
  The Apache License 2.0 is a permissive free software license.
  It allows anyone to use, modify, and distribute the software
  for any purpose, including commercial use, without requiring
  derivative works to be open-sourced (unlike the GPL).

============================================================
  Learning Linux step by step is really helpful.
============================================================

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$
```

Output showing system information

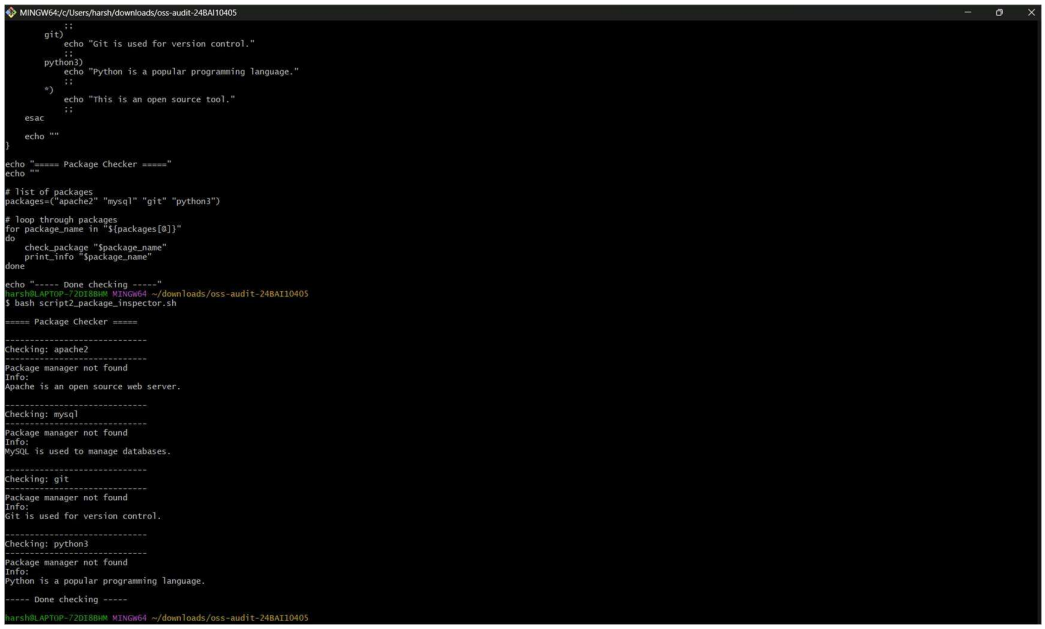
Script 2 — FOSS Package Inspector

This script checks whether a specific package is installed on the system. It uses conditional statements and package management commands such as dpkg to verify the presence of the package. If the package is installed, it displays relevant details. This script demonstrates the use of conditions and package handling in Linux.



```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ cat script2_package_inspector.sh
#!/bin/bash

# this script checks installed packages and shows basic info

# checking which package manager is available
if command -v dpkg >/dev/null
then
    package_manager="dpkg"
elif command -v rpm >/dev/null
then
    package_manager="rpm"
else
    package_manager="unknown"
fi

# function to check package
check_package() {
    package_name=$1

    echo "-----------------------------"
    echo "Checking: $package_name"
    echo "-----------------------------"

    if [ "$package_manager" = "dpkg" ]
    then
        # for ubuntu/debian
        if dpkg -l | grep -q "$package_name"
        then
            echo "Status: Installed"
        else
            echo "Status: Not Installed"
        fi

    elif [ "$package_manager" = "rpm" ]
    then
        # for centos/rhel
        if rpm -q "$package_name" >/dev/null
        then
            echo "Status: Installed"
        else
            echo "Status: Not Installed"
        fi
    else
        echo "Package manager not found"
    fi
}

# simple explanation for each package
print_info() {
    package_name=$1

    echo "Info:"

    case $package_name in
        apache2)
            echo "Apache is an open source web server."
            ;;
        mysql)
            echo "MySQL is used to manage databases."
            ;;
        git)
            echo "Git is used for version control."
            ;;
```

Script to check installed packages



```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405
            ;;
        git)
            echo "Git is used for version control."
            ;;
        python3)
            echo "Python is a popular programming language."
            ;;
        *)
            echo "This is an open source tool."
            ;;
    esac

    echo ""
}

echo "===== Package Checker ====="
echo ""

# list of packages
packages=("apache2" "mysql" "git" "python3")

# loop through packages
for package_name in "${packages[@]}"
do
    check_package "$package_name"
    print_info "$package_name"
done

echo "----- Done checking -----"
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script2_package_inspector.sh

===== Package Checker =====

-----------------------------
Checking: apache2
-----------------------------
Package manager not found
Info:
Apache is an open source web server.

-----------------------------
Checking: mysql
-----------------------------
Package manager not found
Info:
MySQL is used to manage databases.

-----------------------------
Checking: git
-----------------------------
Package manager not found
Info:
Git is used for version control.

-----------------------------
Checking: python3
-----------------------------
Package manager not found
Info:
Python is a popular programming language.

----- Done checking -----

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
```

Output showing package status

Script 3 — Disk and Permission Auditor

This script analyzes important directories in the system. It checks whether directories exist and displays details such as permissions, ownership, and disk usage. It uses loops and commands like ls and du. This script helps in understanding file system structure and permissions in Linux.

```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405
Python is a popular programming language.

----- Done checking -----

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ cat script3_disk_permission_auditor.sh
#!/bin/bash

# I made this script to check directory permissions and size

# basic system directories
directories=("/etc" "/var/log" "/home" "/tmp")

# apache related folders
apache_dirs=("/etc/apache2" "/var/log/apache2" "/var/www/html")

# function to check directory
check_directory() {
    dir=$1

    echo "----------------------------"
    echo "Checking directory: $dir"
    echo "----------------------------"

    # check if directory exists
    if [ -d "$dir" ]
    then
        # show permissions
        echo "Permissions:"
        ls -ld "$dir"

        # show size
        echo "Size:"
        du -sh "$dir" 2>/dev/null
    else
        echo "Directory not found"
    fi

    echo ""
}

echo "Checking system directories"
echo ""

# loop for system directories
for dir in "${directories[@]}"
do
    check_directory "$dir"
done

echo "Checking apache related directories"
echo ""

# loop for apache directories
for dir in "${apache_dirs[@]}"
do
    check_directory "$dir"
done

echo "----- Directory check complete -----"
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$
```

Script for auditing disk usage and permissions

```
MINGW64:/c/Users/harsh/downloads/oss-audit-24BAI10405
do
    check_directory "$dir"
done

echo "Checking apache related directories"
echo ""

# loop for apache directories
for dir in "${apache_dirs[@]}"
do
    check_directory "$dir"
done

echo "----- Directory check complete -----"
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script3_disk_permission_auditor.sh
Checking system directories

----------------------------
Checking directory: /etc
----------------------------
Permissions:
drwxr-xr-x 1 harsh 197609 0 Feb  4 01:09 /etc
Size:
1.6M    /etc

----------------------------
Checking directory: /var/log
----------------------------
Directory not found

----------------------------
Checking directory: /home
----------------------------
Directory not found

----------------------------
Checking directory: /tmp
----------------------------
Permissions:
drwxr-xr-x 1 harsh 197609 0 Mar 31 01:32 /tmp
Size:
728M    /tmp

Checking apache related directories

----------------------------
Checking directory: /etc/apache2
----------------------------
Directory not found

----------------------------
Checking directory: /var/log/apache2
----------------------------
Directory not found

----------------------------
Checking directory: /var/www/html
----------------------------
Directory not found

----- Directory check complete -----

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$
```

Output showing directory details

Script 4 — Log File Analyzer

This script reads a log file and searches for a specific keyword. It counts the number of matching lines and displays recent entries. It uses loops and commands like grep and tail for processing text. This script demonstrates how log

```
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ cat script4_log_analyzer.sh
#!/bin/bash

# I made this script to check logs and find keyword occurrences

# take log file from user or use default
log_file=${1:-"/var/log/syslog"}

# take keyword or use default "error"
keyword=${2:-"error"}

# check if file exists
if [ ! -f "$log_file" ]
then
    echo "Log file not found"
    exit 1
fi

echo "Checking log file: $log_file"
echo "Searching for: $keyword"
echo ""

# count number of matches
match_count=$(grep -i "$keyword" "$log_file" | wc -l)

echo "Total matches found: $match_count"
echo ""

echo "Last 5 matching lines:"
grep -i "$keyword" "$log_file" | tail -5

echo ""
echo "----- Log check complete -----"

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
```

files can be analyzed using shell scripting.

```
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script4_log_analyzer.sh
Log file not found

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ ^C

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ touch sample.log

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ echo "error: login failed" >> sample.log
echo "info: system running" >> sample.log
echo "error: disk full" >> sample.log

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script4_log_analyzer.sh sample.log
Checking log file: sample.log
Searching for: error

Total matches found: 2

Last 5 matching lines:
error: login failed
error: disk full

----- Log check complete -----

harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$
```

The script initially displayed an error due to the absence of the log file. A sample log file was created and used as input to successfully execute the script and analyze log entries.

Script 5 — Open Source Manifesto Generator

The script was executed successfully, and user input was provided for all questions. Based on the responses, a personalized manifesto was generated and saved into a text file. The output was also displayed on the terminal.

```
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ cat script5_manifesto_generator.sh
#!/bin/bash

# I made this script to take user input and create a simple manifesto file

# output file name
file_name="my_manifesto.txt"

echo "Open Source Manifesto Generator"
echo ""

# taking input from user
read -p "Enter a tool you use: " tool
read -p "What does freedom mean to you? " freedom
read -p "What would you like to build? " idea

# if user leaves input empty, give default values
[ -z "$tool" ] && tool="Linux"
[ -z "$freedom" ] && freedom="sharing"
[ -z "$idea" ] && idea="a helpful tool"

# writing data into file
echo "----- My Open Source Manifesto -----" > "$file_name"
echo "User: $(whoami)" >> "$file_name"
echo "Date: $(date)" >> "$file_name"
echo "" >> "$file_name"

echo "I use $tool in my daily work." >> "$file_name"
echo "For me, freedom means $freedom." >> "$file_name"
echo "I would like to build $idea and share it with others." >> "$file_name"

echo ""
echo "File created: $file_name"
echo ""

# display file content
cat "$file_name"

echo ""
echo "----- Done -----"
```

```
harsh@LAPTOP-72DI8BHM MINGW64 ~/downloads/oss-audit-24BAI10405
$ bash script5_manifesto_generator.sh
Open Source Manifesto Generator

Enter a tool you use: Git
What does freedom mean to you? Because it is free and flexible
What would you like to build? an app

File created: my_manifesto.txt

----- My Open Source Manifesto -----
User: harsh
Date: Tue Mar 31 01:53:40 IST 2026

I use Git in my daily work.
For me, freedom means Because it is free and flexible.
I would like to build an app and share it with others.

----- Done -----
```

The script was executed successfully, and user input was provided for each prompt. Based on the responses, a personalized open-source manifesto was generated and displayed on the terminal. The output was also saved to a text file for future reference

CONCLUSION

This project gave me hands-on experience with open-source technologies and Linux systems. I learned how tools like Git and the Apache web server work.

I also gained experience in writing shell scripts to automate tasks and analyze systems. The scripts I created showed me how to get system information manage packages handle files and analyze logs. Doing these tasks helped me improve my problem-solving skills.

I got to practice with real-world system operations. Using open-source tools like Git is really helpful. They are flexible transparent. Do not cost much. Students and developers like me can benefit from them. This project helped me understand Linux environments better.

It also improved my skills in writing shell scripts and using open-source software. I feel more confident now in working with Linux and open-source tools, like Git..